

Symbiosis Institute Of Technology, Pune



Department Of Applied Science
First Year B. Tech Academic Year: 2025-26

Python Lab Project Report

Comprehensive Analysis and Visualization of Indian Car Sales Trends

Project By:

1. Maahi Sadrani - (25070123069)
2. Madhur Gupta - (25070123070)
3. Pranav Menon - (25070123085)
4. Rachit Jajoo - (25070123088)

Faculty In Charge:
Mr. Anurag Mahajan

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our respected faculty guide, **Mr. Anurag Mahajan**, for his valuable guidance, continuous support, and encouragement throughout the development of this project titled “**Comprehensive Analysis and Visualization of Indian Car Sales Trends.**” His insightful suggestions, technical expertise, and motivation played a significant role in the successful completion of this work.

We are also thankful to the **Department of Applied Science, Symbiosis Institute of Technology, Pune**, for providing us with the necessary resources, facilities, and a supportive academic environment required to carry out this project successfully.

We would like to extend our appreciation to all our team members for their dedication, cooperation, and collective efforts during every phase of the project, including data collection, analysis, model development, report preparation, and website implementation.

We also acknowledge the official documentation, online references, and technical resources related to Python programming, machine learning, and web development that guided us throughout the execution of this project.

Finally, we express our heartfelt gratitude to our parents, friends, and well-wishers for their constant encouragement and moral support during the completion of this project.

Their assistance and motivation were invaluable in making this project a success.

INTRODUCTION

The Indian automobile industry is one of the most important and rapidly growing sectors of the national economy. It contributes significantly to industrial development, employment generation, technological advancement, and consumer mobility. With increasing competition among automobile companies and changing customer preferences, the use of data-driven analysis has become essential for understanding market trends and forecasting future demand.

This project, titled “**Comprehensive Analysis and Visualization of Indian Car Sales Trends,**” presents a comparative study of three leading automobile manufacturers in India, namely **Maruti Suzuki, Mahindra, and Hyundai**. These brands represent a major share of the Indian passenger vehicle market and provide an ideal foundation for analytical study. The project uses structured datasets containing variables such as units sold, ex-showroom price, revenue, mileage, fuel type, transmission type, customer rating, safety rating, seating capacity, annual service cost, and time-based attributes.

A systematic approach has been followed involving data collection, cleaning, preprocessing, transformation, and integration of multiple datasets into a single analytical framework. The concepts of **Exploratory Data Analysis (EDA)** have been applied to identify patterns, detect anomalies, and understand relationships between variables. Various visualizations such as boxplots, heatmaps, bar charts, line graphs, stacked charts, and bubble charts have been used to represent the data effectively.

The project also includes **Machine Learning** for predictive analysis. A **Random Forest Regressor** model has been implemented to estimate future sales based on multiple automobile and market-related features. This enables the project to move beyond historical analysis and provide intelligent forecasting insights.

To enhance practical usability, an interactive web dashboard named **AutoIntel** has been developed using HTML, CSS, and JavaScript. The platform allows users to explore datasets, compare brand performance, visualize trends, and generate forecasts through a modern user interface.

Overall, this project demonstrates the effective integration of data analytics, machine learning, and web technologies to study the Indian automobile market and support informed business decision-making.

OBJECTIVE OF THE PROJECT

1. To analyze the sales performance of leading Indian automobile brands:

- a. Study the historical sales data of **Maruti Suzuki, Mahindra, and Hyundai** to understand their market performance, growth trends, and comparative position in the Indian automobile sector.

2. To apply Exploratory Data Analysis (EDA):

- a. Use statistical methods and analytical techniques to identify patterns, trends, relationships, and anomalies present in automobile sales data.

3. To create meaningful graphical visualizations:

- a. Develop charts such as boxplots, heatmaps, bar graphs, line charts, stacked charts, and bubble charts for clear representation and better understanding of the data.

4. To compare different brands and market segments:

- a. Examine brand-wise performance based on factors such as pricing, mileage, customer ratings, fuel type, safety ratings, revenue, and units sold.

5. To derive useful business insights from the dataset:

- a. Interpret the results obtained from analysis and visualization to understand customer preferences, market behaviour, and sales trends.

RAW DATA

The foundation of every successful data analytics project lies in the quality and relevance of the dataset used. For this project, structured sales datasets of three major automobile manufacturers in India were selected to perform a comparative study of market trends and brand performance.

The datasets used are:

1. **Maruti Suzuki.csv**
2. **Mahindra.csv**
3. **Hyundai.csv**

These companies were chosen because they represent a significant share of the Indian passenger vehicle market and cater to diverse customer segments including hatchbacks, sedans, SUVs, and utility vehicles.

After performing data cleaning, preprocessing, and integration, the three datasets were merged into a single master dataset containing **2204 records** for detailed analysis.

The data covers the period from **2020 to 2024**, enabling the study of historical sales patterns, seasonal variations, yearly growth trends, and comparative brand performance over multiple years.

Important Attributes Included in the Dataset:

- Month
- Year
- Brand
- Car Model
- Segment
- Fuel Type
- Transmission
- Ex-Showroom Price (Lakh)
- Units Sold
- Revenue (Crores)
- Mileage (KMPL)
- Customer Rating
- Annual Service Cost (INR)
- Safety Rating (Stars)
- Seating Capacity
- Quarter

The dataset contains both numerical and categorical variables, making it highly suitable for statistical analysis, data visualization, and trend identification.

Thus, the selected raw data provides a comprehensive representation of the Indian automobile market and forms a strong basis for Exploratory Data Analysis, comparative evaluation, and meaningful business insights.

Here are some references to the Raw Data:

#	A	B	C	D	E	F	G	H	I	J	K	L
1	Month	Car_Model	Segment	Fuel_Type	Transmission	Ex_Showroom_Price_Lakh	Units_Sold	Revenue_Crores	Mileage_KMPL	Customer_Rating	Annual_Service_Cost_INR	Safety_Rating_Stars
2	2020-01	Alto K10	Hatchback	Petrol	Manual	4.54	7623	346.08	24.54	5	12566	3
3	2020-01	Baleno	Hatchback	Petrol	Manual/AMT	7.61	7583	577.07	23.34	4	8357	5
4	2020-01	Brezza	SUV	Petrol/CNG	Manual/AT	8.1	5743	465.18	19.4	4.4	12557	5
5	2020-01	Celerio	Hatchback	Petrol/CNG	Manual/AMT	5.28	8014	423.14	25.83	4.3	8107	3
6	2020-01	Ciaz	Sedan	Petrol	Manual	8.76	3998	341.46	20.43	5	11131	5
7	2020-01	Dzire	Sedan	Petrol	Manual/AMT	6.73	4011	269.94	22.64	4.5	11623	4
8	2020-01	Eeco	MPV	Petrol/CNG	Manual	5.4	4358	235.33	19.09	3.9	10381	4
9	2020-01	Ertega	MPV	Petrol/CNG	Manual/AT	8.43	4196	353.72	20.27	4.5	13399	5
10	2020-01	Fronx	SUV	Petrol/CNG	Manual/AMT/AT	8.49	5990	508.55	21.91	4.7	6589	4
11	2020-01	Grand Vitara	SUV	Petrol/Strong Hybrid	Manual/AT	10.44	6724	701.99	21.86	3.8	9969	5
12	2020-01	S-Presso	Hatchback	Petrol/CNG	Manual/AMT	4.7	8931	419.76	25.21	5	11413	4
13	2020-01	Swift	Hatchback	Petrol	Manual/AMT	7.08	7940	562.15	22.85	3.9	6160	3
14	2020-01	WagonR	Hatchback	Petrol/CNG	Manual/AMT	5.35	7475	399.91	24.09	4.4	10374	3
15	2020-01	XL6	MPV	Petrol/CNG	Manual	11.9	4161	495.16	20.79	4.5	9061	5
16	2020-02	Alto K10	Hatchback	Petrol	Manual	4.41	7556	333.22	23.57	4.1	9287	3
17	2020-02	Baleno	Hatchback	Petrol	Manual/AMT	7.45	9304	693.15	23.18	4.2	7058	5
18	2020-02	Brezza	SUV	Petrol/CNG	Manual/AT	9.63	7517	723.89	19.88	4.4	11915	5
19	2020-02	Celerio	Hatchback	Petrol/CNG	Manual/AMT	5.51	8451	465.65	26.15	4.8	11695	3
20	2020-02	Dzire	Sedan	Petrol	Manual/AMT	7.24	3637	263.32	23.66	4.7	9619	4
21	2020-02	Eeco	MPV	Petrol/CNG	Manual	5	8669	229.57	20.47	3.7	5624	5
22	2020-02	Ertega	MPV	Petrol/CNG	Manual/AT	9.72	4172	405.52	20.79	4.9	10444	5
23	2020-02	Fronx	SUV	Petrol/CNG	Manual/AMT/AT	9.3	6456	600.41	21.41	4.4	9437	4
24	2020-02	Grand Vitara	SUV	Petrol/Strong Hybrid	Manual/AT	11.18	7390	826.2	21.33	4.6	8052	5
25	2020-02	S-Presso	Hatchback	Petrol/CNG	Manual/AMT	4.92	7358	362.01	25.88	4.6	10889	4
26	2020-02	Swift	Hatchback	Petrol	Manual/AMT	6.86	7124	488.71	23.36	4.5	8780	3
27	2020-02	WagonR	Hatchback	Petrol/CNG	Manual/AMT	5.58	8710	486.02	23.51	5	8084	3
28	2020-03	Alto K10	Hatchback	Petrol	Manual	4.669	280.1	240.0	4.7		6917	5
29	2020-03	Baleno	Hatchback	Petrol	Manual/AMT	7.44	7280	541.63	23.17	4.7	10047	5
30	2020-03	Brezza	SUV	Petrol/CNG	Manual/AT	10.51	6690	703.12	19.18	3.9	10666	5
31	2020-03	Celerio	Hatchback	Petrol/CNG	Manual/AMT	6.25	8318	519.88	27.42	4.4	9408	3
32	2020-03	Ciaz	Sedan	Petrol	Manual	8.71	3913	340.82	20.46	4.3	10901	5
33	2020-03	Dzire	Sedan	Petrol	Manual/AMT	6.57	3811	250.38	23.92	5	10320	4
34	2020-03	Eeco	MPV	Petrol/CNG	Manual	5.29	4611	243.92	20.01	3.9	4606	4
35	2020-03	Ertega	MPV	Petrol/CNG	Manual/AMT	9.72	3627	371.98	20.62	3.9	14701	5
36	2020-03	Fronx	SUV	Petrol/CNG	Manual/AMT/AT	8.76	5745	503.26	22.62	4.5	13160	4
37	2020-03	Grand Vitara	SUV	Petrol/Strong Hybrid	Manual/AT	13.88	5248	728.42	20.76	4.1	14218	5
38	2020-03	S-Presso	Hatchback	Petrol/CNG	Manual/AMT	3.71	7222	267.94	25.06	3.8	10583	4
39	2020-03	Swift	Hatchback	Petrol	Manual/AMT	7.29	9062	660.62	23.09	4.3	11179	3
40	2020-03	WagonR	Hatchback	Petrol/CNG	Manual/AMT	5.63	7413	417.35	23.68	4.1	13294	3
41	2020-03	XL6	MPV	Petrol/CNG	Manual/AT	11.05	5313	587.09	19.95	4.2	9454	5
42	2020-04	Baleno	Hatchback	Petrol	Manual/AMT	7.64	8187	625.49	22.81	4.4	10966	5
43	2020-04	Brezza	SUV	Petrol/CNG	Manual/AT	9.97	5744	509.46	20.44	4.5	12123	5

#	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Month	Car_Model	Segment	Fuel_Type	Engine_CC	Transmission	Seating_Capacity	Ex_Showroom_Price_Lakh	Units_Sold	Revenue_Crores	Mileage_KMPL	Customer_Rating	Annual_Service_Cost_INR	Warranty_Years	Safety_Rating_Stars
2	2023-02	Thar	SUV	Petrol	2184	Automatic	7	12.85	6206	10647.51	17.7	3.9	5391	5	4
3	2023-02	Thar	SUV	Petrol	1497	Automatic	5	23.4	6164	14423.76	16.5	4.1	10527	5	3
4	2024-09	Bolero	SUV	Diesel	2184	Manual	7	19.79	10737	21248.52	19	4.2	10182	5	5
5	2024-12	Marazzo	MPV	Petrol	1197	Automatic	5	6.12	3296	2017.15	16.4	3.7	10942	3	4
6	2023-06	KUV100	Hatchback	Diesel	1997	Automatic	7	15.8	2862	4521.96	14.2	3.6	11949	3	4
7	2024-01	KUV300	SUV	Diesel	2184	Manual	5	22.4	7650	17136	17.9	3.7	9733	5	3
8	2024-02	Marazzo	MPV	Diesel	2184	Automatic	5	10.39	10628	11250.29	15.3	4.2	7783	3	4
9	2024-01	Scorpio	SUV	Petrol	1997	Manual	5	12.71	12249	15568.48	16	4.1	8986	3	3
10	2023-01	KUV700	SUV	Petrol	2184	Automatic	7	17.13	9142	15660.25	16.3	3.6	10734	3	3
11	2023-07	KUV300	SUV	Diesel	1497	Automatic	5	18.11	10860	19667.46	19.5	4	11615	5	3
12	2023-04	Scorpio N	SUV	Petrol	1997	Manual	7	22.92	3130	7173.96	16.4	3.9	11598	3	5
13	2023-02	Marazzo	MPV	Diesel	1197	Automatic	5	18.51	12706	23518.81	18.4	3.6	6388	5	5
14	2024-01	Scorpio N	SUV	Petrol	1997	Manual	5	12.96	6023	7817.85	16.9	3.8	8707	5	4
15	2023-06	Marazzo	MPV	Diesel	1197	Automatic	5	18.81	6947	13067.31	15.1	4.1	6161	3	3
16	2023-05	KUV700	SUV	Diesel	1997	Manual	7	11.86	3729	4422.59	19.4	3.7	10300	3	3
17	2023-07	Scorpio N	SUV	Petrol	1997	Automatic	5	6.68	8291	5538.39	17.4	4.3	9722	5	5
18	2024-04	Marazzo	MPV	Diesel	1497	Manual	7	22.43	3942	8841.91	15.3	4.7	11728	3	3
19	2023-05	Marazzo	MPV	Petrol	1997	Manual	5	12.11	477	577.65	18.8	3.6	11445	5	3
20	2023-11	Scorpio	SUV	Diesel	2184	Manual	5	15.04	7233	10678.43	15.1	3.8	8789	5	4
21	2023-05	KUV700	SUV	Diesel	2184	Automatic	7	22.1	2554	5644.34	19.2	4.4	6208	5	4
22	2024-03	KUV700	SUV	Diesel	1497	Automatic	5	7.16	1700	1217.2	15.8	4.5	9960	5	5
23	2023-02	KUV300	SUV	Petrol	1497	Manual	5	13.31	8834	11758.05	18	4.3	6873	3	3
24	2023-04	Scorpio	MPV	Petrol	1197	Automatic	5	24.67	3384	8348.33	15.4	3.8	6208	3	4
25	2023-11	KUV700	SUV	Diesel	2184	Automatic	5	19.28	11892	22927.78	19.2	4.3	6958	3	4
26	2024-06	Thar	SUV	Diesel	2184	Manual	5	23.62	8691	20528.14	15.5	4	6926	3	4
27	2024-06	Marazzo	MPV	Diesel	1997	Manual	5	16.33	5161	8427.91	19.4	3.7	12832	3	3
28	2023-01	Thar	SUV	Diesel	1197	Automatic	7	21.62	3659	8343.16	14.6	4.6	8797	3	3
29	2023-12	KUV300	SUV	Diesel	1197	Automatic	5	11.64	3316	3859.82	14.4	4.3	9460	3	3
30	2024-08	KUV300	SUV	Petrol	1997	Manual	7	17.78	10739	19093.94	19.5	3.7	7721	5	5
31	2023-09	KUV300	SUV	Petrol	1997	Automatic	5	6.34	10896	6908.06	16.2	3.5	7517	5	3
32	2024-02	Scorpio	SUV	Diesel	1197	Automatic	5	10.34	12650	13080.1	15.4	3.5	12795	3	5
33	2024-08	Scorpio	SUV	Diesel	1497	Manual	5	9.38	3550	3329.9	16.3	4.5	6395	5	3
34	2024-02	KUV100	Hatchback	Diesel	1997	Automatic	5	21.69	9294	20158.69	17.4	3.8	10482	5	5
35	2024-12	Scorpio N	SUV	Diesel	1497	Automatic	7	6.24	8816	5501.18	18.1	4.6	8626	3	3
36	2023-09	Scorpio N	SUV	Diesel	1997	Manual	7	21.86	12075	26395.95	19.3	4.4	8456	5	3
37	2023-09	Marazzo	MPV	Diesel	1997	Automatic	5	11.23	4999	5613.88	19.4	3.7	11298	5	5
38	2024-09	Scorpio N	SUV	Petrol	1197	Manual	5	12.1	11790	14265.9	15.1	3.6	7716	3	3

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Month	Car_Model	Segment	Fuel_Type	Engine_CC	Transmission	Seating_Capacity	Ex_Showroom_Price_Lakh	Units_Sold	Revenue_Crores	Mileage_KMPL	Customer_Rating	Annual_Service_Cost_INR	Warranty_Years	Safety_Rating_Stars
2	2023-02	Venue	SUV	Diesel	1197	Automatic	5	18.16	13593	24684.89	21.3	3.9	11092	5	5
3	2024-07	Aura	Sedan	Petrol	1197	Manual	5	20.73	1911	3961.5	19.3	4.4	5069	3	3
4	2024-11	Exter	SUV	Diesel	1497	Automatic	5	10.55	7739	8164.65	18.9	4	10907	3	5
5	2024-06	Aura	Sedan	Petrol	999	Manual	5	6.64	6192	4111.49	17.4	3.6	5562	5	4
6	2024-11	Grand i10 Nio	Hatchback	Petrol	999	Manual	5	7.96	2961	2356.96	19.5	4.1	10194	5	3
7	2023-10	Aura	Sedan	Petrol	999	Automatic	7	13.87	5983	8298.42	17.2	4.5	10306	5	4
8	2023-01	Venue	SUV	Diesel	999	Manual	5	21.5	6625	14243.75	20.9	4.5	4762	3	5
9	2024-04	Creta	SUV	Diesel	1999	Automatic	5	17.25	5967	10293.08	17.9	4.7	9451	5	4
10	2023-07	Tucson	SUV	Diesel	1497	Manual	5	6.92	5573	3856.52	15.2	4.3	9732	3	3
11	2023-11	Tucson	SUV	Diesel	1999	Automatic	5	19.99	1023	2044.98	20.8	3.9	6470	5	4
12	2024-01	Tucson	SUV	Diesel	1497	Manual	5	15	4713	7069.5	20.9	4.5	4923	5	3
13	2024-01	Tucson	SUV	Petrol	1197	Automatic	5	22.79	12805	29182.6	21.3	3.7	5521	5	4
14	2024-05	Tucson	SUV	Diesel	1497	Automatic	5	24.75	6301	15594.98	16.3	4.2	7927	5	4
15	2024-11	Tucson	SUV	Petrol	1497	Automatic	5	20.37	2174	4428.44	22	3.9	7259	5	4
16	2024-12	Tucson	SUV	Petrol	1497	Automatic	5	6	13906	8343.6	16.8	4.3	8809	5	3
17	2024-03	Alcazar	SUV	Petrol	1999	Manual	5	17.2	13689	23545.08	20.7	4.3	6587	5	5
18	2024-06	Grand i10 Nio	Hatchback	Diesel	1197	Manual	5	23.91	7268	5422.79	15.2	4.8	6136	3	3
19	2023-12	Verna	Sedan	Petrol	1197	Automatic	5	7.9	6051	4780.29	17.7	4.8	8415	3	5
20	2023-01	Venue	SUV	Petrol	1197	Manual	5	14.33	8578	12292.27	19	3.7	11420	5	5
21	2024-07	Verna	Sedan	Petrol	999	Automatic	5	20.68	4659	9634.81	16.7	4.6	6541	5	5
22	2023-06	Alcazar	SUV	Diesel	1999	Manual	7	6.44	12353	7955.33	19.5	4.6	7318	3	3
23	2024-01	Creta	SUV	Diesel	1197	Manual	7	7.57	7037	5327.01	21	4.2	6891	5	3
24	2023-09	Exter	SUV	Diesel	999	Automatic	5	23.51	1634	3841.53	20.2	3.5	6916	5	4
25	2024-02	Verna	Sedan	Petrol	1497	Manual	5	17.85	9482	16925.37	20.5	4.7	5039	3	3
26	2023-08	Venue	SUV	Diesel	999	Manual	5	8.5	4232	3597.2	16.6	3.7	7196	5	5
27	2024-06	i20	Hatchback	Petrol	999	Manual	5	13.07	9922	12968.05	21.8	3.7	11024	5	5
28	2024-06	Aura	Sedan	Diesel	1197	Automatic	5	21.26	6202	13185.45	16.5	3.9	9714	5	4
29	2024-09	Alcazar	SUV	Petrol	1197	Automatic	5	9.29	5616	5217.26	19	3.9	6172	3	5
30	2023-08	Verna	Sedan	Petrol	1999	Automatic	5	9.96	8427	8393.29	15.3	4.2	4263	3	3
31	2024-08	Tucson	SUV	Petrol	1197	Manual	5	11.87	2647	3141.99	19	4.1	8962	3	4
32	2024-12	Tucson	SUV	Diesel	1197	Automatic	5	13.79	2557	3526.1	18.7	4.4	4990	5	3
33	2024-01	Tucson	SUV	Petrol	1497	Automatic	5	16.39	12231	20046.61	20	4.5	6193	5	4
34	>	hyundai_car_sales (1)													

FUNCTIONS USED

Python offers a wide range of libraries that make data analysis, visualization, and machine learning more efficient and effective. In this project, several libraries and web technologies were used to process the datasets, generate meaningful insights, and build an interactive dashboard.

The major tools used in the project are as follows:

1. Pandas

Pandas was used for data handling and preprocessing. It helped in reading CSV files, cleaning the datasets, removing missing values, combining multiple datasets, and performing structured data analysis.

Important Functions Used:

`read_csv()`, `dropna()`, `drop_duplicates()`, `groupby()`, `concat()`, `reset_index()`

2. NumPy

NumPy was used for numerical computations and statistical operations. It supported calculations such as averages, standard deviation, variance, and other mathematical processes required during analysis.

Important Functions Used:

`mean()`, `std()`, `var()`, `corrcoef()`

3. Matplotlib

Matplotlib was used to create various graphs and charts for visual representation of the automobile data. It was useful for generating line charts, bar graphs, scatter plots, and boxplots.

Important Functions Used:

`plot()`, `bar()`, `scatter()`, `boxplot()`, `show()`

4. Seaborn

Seaborn was used to create more advanced and visually appealing statistical charts. It was mainly used for correlation heatmaps and comparative graphical analysis.

Important Functions Used:

`heatmap()`, `boxplot()`

5. Scikit-learn

Scikit-learn was used for implementing machine learning techniques. In this project, it was used to build the **Random Forest Regressor** model for predicting future sales trends.

Important Functions Used:

`train_test_split()`, `RandomForestRegressor()`, `fit()`, `predict()`, `r2_score()`

6. HTML, CSS and Python Code

These technologies were used to design the web dashboard **AutoIntel**, which provides an interactive platform for viewing graphs, comparing brands, and exploring project results.

THEORY

Data analysis is the systematic process of collecting, organizing, cleaning, and interpreting data in order to derive meaningful insights and support informed decision-making. In today's competitive business environment, data has become one of the most valuable assets for understanding customer preferences, identifying market trends, measuring performance, and planning future growth strategies.

In this project, the sales data of major Indian automobile companies was analyzed using Python-based tools and modern analytical techniques. The important theoretical concepts applied in the project are explained below:

1. Data Cleaning

Real-world datasets often contain missing values, duplicate records, inconsistent formats, and invalid entries. Data cleaning is the process of identifying and correcting such issues so that the dataset becomes accurate, consistent, and suitable for analysis.

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is used to study the overall structure of the dataset and uncover hidden patterns, trends, and relationships between variables. It combines statistical summaries with visual methods to improve understanding of the data before detailed modelling.

3. Data Visualization

Data visualization refers to the graphical representation of data using charts, graphs, and plots. It transforms complex numerical information into an easy-to-understand format, helping users identify trends, comparisons, and unusual observations quickly.

4. Comparative Analysis And Correlation Analysis

Comparative analysis is used to evaluate multiple entities using common parameters, while correlation analysis measures the relationship between numerical variables. In this project, Maruti Suzuki, Mahindra, and Hyundai were compared on the basis of sales, pricing, mileage, customer ratings, and revenue. Correlation analysis was also used to study the relationship between variables affecting sales trends.

5. Machine Learning

Machine Learning enables computer systems to learn patterns from historical data and make future predictions. As an advanced extension of this project, a Random Forest Regressor model was implemented to estimate future automobile sales trends.

ALGORITHMS OF THE PROJECT

The project was carried out in a systematic manner to analyze the sales trends of major Indian automobile brands. Here is an overall outline we followed during the execution of the project:

1. Start the program and import all required Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.
2. Load the datasets of **Maruti Suzuki**, **Mahindra**, and **Hyundai** from CSV files.
3. Inspect the datasets and identify missing values, duplicate records, and formatting inconsistencies.
4. Perform data cleaning by removing null values, duplicate entries, and correcting column formats.
5. Convert required columns into appropriate numerical and date formats for analysis.
6. Merge the cleaned datasets into a single master dataset and add brand labels.
7. Create additional attributes such as Year, Quarter, Month Number, and Price Band for better analysis.
8. Apply Exploratory Data Analysis (EDA) techniques to calculate summary statistics and study patterns in the data.
9. Generate visualizations such as bar charts, line graphs, boxplots, heatmaps, and bubble charts for comparative analysis.
10. Compare the performance of different brands based on sales, revenue, mileage, ratings, and other parameters.
11. Implement the Random Forest Regressor model as an advanced step to estimate future sales trends.
12. Develop the **AutoIntel** dashboard to present insights, graphs, and forecasts in an interactive format.
13. Display the final results and conclusions obtained from the analysis.
14. Stop.

This was the main algorithm outline for the entire project.

However, the project was implemented through multiple stages including data preprocessing, statistical analysis, visualization, and machine learning. The important algorithms used in the project codes are given below.

Algorithm 1: Data Cleaning

1. Start the program.
2. Import required Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.
3. Read the raw CSV dataset.
4. Display the first few records using `head()`.
5. Check dataset structure using `info()`.
6. Identify missing values using `isnull().sum()`.
7. Remove duplicate records using `drop_duplicates()`.
8. Remove null values using `dropna()`.
9. Remove extra spaces from column names using `str.strip()`.
10. Convert Month column into datetime format.
11. Convert numerical columns into numeric datatypes.
12. Reset index, verify cleaned data, and save dataset.
13. Stop.

Algorithm 2: Brand-wise Dataset Processing

1. Load Maruti Suzuki dataset.
2. Inspect dataset records and structure.
3. Apply a complete cleaning pipeline.
4. Convert required columns into proper format.
5. Save cleaned Maruti dataset.
6. Load Hyundai dataset.
7. Inspect dataset records and structure.
8. Apply a complete cleaning pipeline.
9. Save cleaned Hyundai dataset.
10. Load Mahindra dataset.
11. Inspect dataset records and structure.
12. Apply a complete cleaning pipeline.
13. Save cleaned Mahindra dataset.
14. Stop.

Algorithm 3: Combined Master Dataset Generation

1. Load all cleaned datasets.
2. Add a Brand column to each dataset.
3. Standardize common columns across all datasets.
4. Merge all datasets using `concat()`.
5. Convert Month column into datetime format.

6. Create a Year column.
7. Create a Quarter column.
8. Create a Month Number column.
9. Create Price Band categories using price ranges.
10. Verify merged dataset dimensions.
11. Store final master dataset.
12. Stop.

Algorithm 4: Univariate Statistical Analysis

1. Select Units Sold variable.
2. Calculate Mean.
3. Calculate Median.
4. Calculate Mode.
5. Calculate Standard Deviation.
6. Calculate Variance.
7. Find Minimum value.
8. Find Maximum value.
9. Calculate the first quartile and third quartile.
10. Detects outliers using the IQR method.
11. Display statistical summary.
12. Stop.

Algorithm 5: Bivariate and Multivariate Analysis

1. Select important variables such as Price, Units Sold, Revenue, Mileage, Rating, and Service Cost.
2. Calculate Pearson correlation coefficients.
3. Compare Price with Units Sold.
4. Compare Mileage with Customer Rating.
5. Compare Customer Rating with Units Sold.
6. Generate correlation matrix.
7. Generate covariance matrix.
8. Create Brand vs Transmission cross-tabulation.
9. Create Brand vs Fuel Type cross-tabulation.
10. Study relationships among multiple variables.
11. Display analytical results.
12. Stop.

Algorithm 6: Data Type Classification and Transformation

1. Identify qualitative and quantitative variables.
2. Classify data as nominal, ordinal, discrete, or continuous.
3. Select important numerical columns.
4. Apply Min-Max Normalization.
5. Apply Z-Score Standardization.
6. Apply Decimal Scaling.
7. Create Price bins using `cut()`.
8. Create Mileage bins using `qcut()`.
9. Create Rating categories.
10. Store transformed dataset.
11. Stop.

Algorithm 7: Categorical Encoding Process

1. Select categorical columns from the dataset.
2. Identify Brand column for Label Encoding.
3. Convert Brand categories into numeric labels.
4. Identify Fuel Type column for Label Encoding.
5. Convert Fuel Type categories into numeric labels.
6. Apply One-Hot Encoding to the Segment column.
7. Generate separate binary columns for each segment.
8. Apply Dummy Encoding to the Transmission column.
9. Remove redundant category columns where required.
10. Merge encoded values with original dataset.
11. Prepare final dataset for machine learning.
12. Stop.

Algorithm 8: Visualization Workflow

1. Load master dataset.
2. Generate Boxplots for spread and outliers by brand.
3. Generate Correlation Heatmap.
4. Generate Segment-wise Units Sold grouped bar chart.
5. Generate Monthly Units Sold trend line graph.
6. Generate Annual Revenue comparison graph.
7. Generate Fuel Type Mix stacked chart.
8. Generate Mileage vs Service Cost bubble chart.
9. Generate Transmission-wise sales chart.
10. Generate Safety Rating distribution graph.

11. Generate Price Band vs Customer Rating graph.
12. Add titles, legends, axis labels, and formatting.
13. Display all graphs.
14. Stop.

Algorithm 9: Machine Learning Prediction using Random Forest

1. Load processed master dataset.
2. Remove rows with missing feature values.
3. Encode categorical variables.
4. Select input features and target variables (Units Sold).
5. Split the dataset into a training set (80%) and a testing set (20%).
6. Initialize Random Forest Regressor with 150 trees.
7. Train model using training data.
8. Predict Units Sold using testing data.
9. Calculate R^2 Score.
10. Calculate RMSE.
11. Calculate MAE.
12. Generate Feature Importance graph.
13. Generate Actual vs Predicted comparison graph.
14. Analyze model performance.
15. Stop.

Algorithm 10: AutoIntel Dashboard Development

1. Design webpage structure using HTML.
2. Create header, navigation bar, and content sections.
3. Apply styling using CSS.
4. Add responsive layout for different screen sizes.
5. Use JavaScript for interactive features.
6. Create sections for datasets, graphs, and predictions.
7. Embed charts and project insights.
8. Add brand comparison modules.
9. Add a prediction display section.
10. Test webpage functionality and layout.
11. Improve user interface for better readability.
12. Launch final AutoIntel dashboard.
13. Stop.

FORMULAS USED

Various statistical, preprocessing, and machine learning formulas were used in this project for data analysis, transformation, visualization, and prediction. The important formulas used are given below:

1. Mean

The mean represents the average value of a dataset.

$$\text{Mean} = \Sigma x / n$$

Where:

$\bar{x}$ = mean

x = individual values

n = total observations

2. Median

The median is the middle value of an ordered dataset.

If n is odd: Middle value

If n is even: (Sum of two middle values) / 2

3. Mode

The mode is the most frequently occurring value.

Mode = Most Frequent Value

4. Range

Range shows the spread of data.

Range = Maximum Value – Minimum Value

5. Variance

Variance measures dispersion of values from the mean.

$$\text{Variance} = \Sigma (x - \bar{x})^2 / n$$

6. Standard Deviation

It is the square root of variance.

$$\text{Standard Deviation} = \sqrt{[\Sigma(x - \bar{x})^2 / n]}$$

7. First Quartile (Q1)

The first quartile represents the 25th percentile.

$$Q1 = (n + 1) / 4 \text{th value}$$

8. Third Quartile (Q3)

The third quartile represents the 75th percentile.

$$Q3 = 3(n + 1) / 4 \text{th value}$$

9. Interquartile Range (IQR)

Used for outlier detection.

$$IQR = Q3 - Q1$$

Outlier Limits:

$$\text{Lower Limit} = Q1 - 1.5(IQR)$$

$$\text{Upper Limit} = Q3 + 1.5(IQR)$$

10. Covariance

Measures joint variation of two variables.

$$\text{Cov}(X,Y) = \Sigma(x - \bar{x})(y - \bar{y}) / n$$

11. Pearson Correlation Coefficient

Measures strength and direction of linear relationship.

$$r = \Sigma(x - \bar{x})(y - \bar{y}) / \sqrt{[\Sigma(x - \bar{x})^2 \times \Sigma(y - \bar{y})^2]}$$

12. Min-Max Normalization

Used to scale values between 0 and 1.

$$\mathbf{X'} = (\mathbf{X} - \mathbf{Xmin}) / (\mathbf{Xmax} - \mathbf{Xmin})$$

13. Z-Score Standardization

Used to transform data with mean 0 and standard deviation 1.

$$\mathbf{Z} = (\mathbf{X} - \boldsymbol{\mu}) / \boldsymbol{\sigma}$$

14. Decimal Scaling

Used to reduce values by powers of 10.

$$\mathbf{X'} = \mathbf{X} / 10^j$$

Where:

j = smallest integer such that $\max(|X'|) < 1$

15. Revenue Formula

Used to estimate revenue generated.

$$\mathbf{Revenue} = \mathbf{Price} \times \mathbf{Units Sold}$$

16. Mean Absolute Error (MAE)

Measures average absolute prediction error.

$$\mathbf{MAE} = (1/n) \sum |y_i - \hat{y}_i|$$

17. Mean Squared Error (MSE)

Measures average squared error.

$$\mathbf{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$$

18. Root Mean Squared Error (RMSE)

Square root of MSE.

$$\mathbf{RMSE} = \sqrt{[(1/n) \sum (y_i - \hat{y}_i)^2]}$$

19. R² Score

Measures goodness of fit of regression model.

$$R^2 = 1 - [\Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2]$$

20. Train-Test Split Ratio

Used for model training and evaluation.

Training Data = 80%

Testing Data = 20%

21. Random Forest Prediction

Final prediction is average of all decision tree outputs.

$$\hat{y} = (1/N) \Sigma T_i(x)$$

Where:

N = number of trees

T_i(x) = output of ith tree

Thus, the above formulas formed the mathematical foundation of the project and were used in statistical analysis, preprocessing, visualization, and machine learning prediction of Indian car sales trends.

COMMANDS USED

Various Python commands were used in this project for data loading, cleaning, preprocessing, analysis, visualization, and machine learning. The important commands used are given below:

1. Data Import Commands

Used to import required libraries.

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import warnings
```

2. Reading Dataset Files

Used to load CSV files into DataFrames.

```
pd.read_csv()
```

Example:

```
ms = pd.read_csv("Maruti Suzuki.csv")
```

```
hy = pd.read_csv("Hyundai.csv")
```

```
mh = pd.read_csv("Mahindra.csv")
```

3. Dataset Inspection Commands

Used to view and understand dataset structure.

```
head()
```

```
info()
```

```
shape
```

```
columns
```

```
dtypes
```

Example:

```
df.head()
```

```
df.info()
```

```
df.shape
```

4. Missing Value Commands

Used to detect and remove null values.

```
isnull().sum()
```

```
dropna()
```

Example:

```
df.isnull().sum()
```

```
df.dropna()
```

5. Duplicate Removal Commands

Used to remove repeated records.

```
drop_duplicates()
```

Example:

```
df.drop_duplicates()
```

6. String Handling Commands

Used to clean column names.

```
str.strip()
```

Example:

```
df.columns = df.columns.str.strip()
```

7. Data Type Conversion Commands

Used to convert columns into correct formats.

```
pd.to_datetime()
```

```
pd.to_numeric()
```

`astype()`

Example:

```
pd.to_datetime(df["Month"])
```

```
pd.to_numeric(df["Units_Sold"])
```

8. Index Management Commands

Used to reset row indexing.

`reset_index()`

Example:

```
df.reset_index(drop=True)
```

9. Saving Cleaned Dataset

Used to save processed files.

`to_csv()`

Example:

```
df.to_csv("cleaned_data.csv", index=False)
```

10. Dataset Merging Commands

Used to combine multiple datasets.

`pd.concat()`

Example:

```
pd.concat([ms, mh, hy], ignore_index=True)
```

11. Date Feature Extraction Commands

Used to extract year, quarter, and month.

`dt.year`

`dt.quarter`

`dt.month`

Example:

```
df["Year"] = df["Month"].dt.year
```

12. Statistical Commands

Used for descriptive statistics.

```
mean()
```

```
median()
```

```
mode()
```

```
std()
```

```
var()
```

```
min()
```

```
max()
```

```
quantile()
```

```
corr()
```

```
cov()
```

Example:

```
df["Units_Sold"].mean()
```

```
df["Units_Sold"].std()
```

13. Crosstab Commands

Used for categorical comparison.

```
pd.crosstab()
```

Example:

```
pd.crosstab(df["Brand"], df["Transmission"])
```

14. Grouping Commands

Used for summary analysis.

```
groupby()
```

`sum()`

`mean()`

`unstack()`

`value_counts()`

Example:

```
df.groupby(["Brand"])["Units_Sold"].sum()
```

15. Normalization Commands

Used for feature scaling.

`MinMaxScaler()`

`StandardScaler()`

`fit_transform()`

Example:

```
scaler = MinMaxScaler()
```

```
scaler.fit_transform(data)
```

16. Binning Commands

Used to convert continuous data into categories.

`pd.cut()`

`pd.qcut()`

Example:

```
pd.cut(df["Price"], bins=4)
```

17. Encoding Commands

Used to convert categorical data into numeric form.

`LabelEncoder()`

`fit_transform()`

`pd.get_dummies()`

Example:

```
le.fit_transform(df["Brand"])
```

```
pd.get_dummies(df["Segment"])
```

18. Visualization Commands

Used to create graphs and charts.

```
plot()
```

```
bar()
```

```
boxplot()
```

```
scatter()
```

```
figure()
```

```
subplots()
```

```
show()
```

```
title()
```

```
xlabel()
```

```
ylabel()
```

```
legend()
```

```
tight_layout()
```

Example:

```
plt.bar(x, y)
```

```
plt.show()
```

19. Seaborn Graph Commands

Used for heatmaps.

```
sns.heatmap()
```

Example:

```
sns.heatmap(corr_matrix)
```

20. Machine Learning Commands

Used for prediction model building.

```
train_test_split()
```

```
RandomForestRegressor()
```

```
fit()
```

```
predict()
```

Example:

```
rf.fit(X_train, y_train)
```

```
pred = rf.predict(X_test)
```

21. Evaluation Commands

Used to check model performance.

```
r2_score()
```

```
mean_squared_error()
```

```
mean_absolute_error()
```

Example:

```
r2_score(y_test, pred)
```

Thus, the above commands were extensively used for successful implementation of data analysis, visualization, and prediction of Indian car sales trends.

RESULTS AND OBSERVATIONS

The project was successfully executed using Python for data preprocessing, statistical analysis, visualization, and machine learning. The major results and observations obtained from the study are given below:

1. Dataset Preparation Result

After cleaning, preprocessing, and integration of all three datasets (**Maruti Suzuki, Mahindra, and Hyundai**), a final master dataset containing **2204 valid records** was prepared for analysis.

2. Statistical Analysis Result

Descriptive statistical analysis was performed on the **Units Sold** variable. The average sales, central tendency, dispersion, minimum value, and maximum value were successfully calculated using mean, median, mode, standard deviation, and variance.

3. Outlier Detection Result

Using the Interquartile Range (IQR) method and boxplot analysis, no major abnormal outliers were observed in the Units Sold dataset, indicating a balanced sales distribution.

4. Brand-wise Performance Result

Comparative analysis showed that **Maruti Suzuki** maintained the highest overall market presence in terms of total sales volume, while **Mahindra** performed strongly in the SUV segment and **Hyundai** showed balanced performance across multiple segments.

5. Segment-wise Observation

Hatchback and mid-range vehicles contributed significantly to total sales, whereas premium and SUV categories generated strong market demand in higher price segments.

6. Fuel Type Observation

Petrol vehicles recorded the highest demand across the combined dataset. Diesel vehicles retained importance mainly in SUVs and utility vehicle segments.

7. Revenue Analysis Result

Although budget vehicles achieved higher unit sales, premium vehicles generated greater revenue due to their higher ex-showroom prices.

8. Correlation Analysis Result

Correlation analysis was performed among variables such as price, units sold, mileage, customer rating, revenue, and service cost. The generated heatmap and correlation matrix helped in understanding relationships between important automobile factors.

9. Visualization Result

Multiple graphs such as boxplots, heatmaps, grouped bar charts, line graphs, stacked charts, bubble charts, and rating comparison charts were successfully generated to present trends and comparisons in a clear visual format.

10. Data Transformation Result

Normalization, standardization, data binning, and categorical encoding techniques were successfully applied, making the dataset suitable for advanced analytics and machine learning.

11. Machine Learning Result

The **Random Forest Regressor** model was implemented for future sales prediction using multiple automobile features. Model performance was evaluated using **R² Score, RMSE, and MAE**, confirming successful execution of predictive analysis.

12. Dashboard Result

The web-based dashboard **AutoIntel** was successfully developed using HTML, CSS, and JavaScript to display project insights, charts, comparisons, and forecast outputs through an interactive interface.

Final Result

Thus, the project successfully demonstrated that modern data analytics and machine learning techniques can be effectively used to study Indian car sales trends, compare brand performance, derive business insights, and estimate future market demand.

CONCLUSION

The project “**Comprehensive Analysis and Visualization of Indian Car Sales Trends**” was successfully completed using Python for data analysis, visualization, and predictive modelling. The main objective of analyzing and comparing the sales trends of leading automobile brands in India was achieved through a structured and data-driven approach.

The datasets of **Maruti Suzuki, Mahindra, and Hyundai** were carefully cleaned, organized, and merged into a unified dataset for detailed study. Statistical techniques were applied to examine sales performance, pricing patterns, customer preferences, revenue trends, and relationships between important variables.

A variety of visualizations such as bar charts, line graphs, heatmaps, boxplots, and comparative plots were created to represent the findings in a clear and meaningful manner. These graphical insights made it easier to understand market behaviour and compare the performance of different brands.

To further strengthen the project, a **Random Forest Regressor** model was implemented for future sales prediction. This highlighted the practical value of machine learning in forecasting demand and supporting business decisions.

In addition, the **AutoIntel Dashboard** was developed to provide an interactive platform for viewing insights, comparisons, charts, and predictive outputs in a user-friendly format.

Overall, this project demonstrated that modern data analytics and machine learning tools can be effectively used to study the Indian automobile market, generate meaningful insights, and support smarter strategic decision-making in the automotive industry.

FUTURE SCOPE

The project “**Comprehensive Analysis and Visualization of Indian Car Sales Trends**” provides a strong foundation for further research, innovation, and real-world business applications. With additional data sources and advanced technologies, the scope of this project can be significantly expanded in the future. Some major possibilities are as follows:

1. Inclusion of Additional Automobile Brands

The study can be extended by incorporating datasets of other leading automobile manufacturers such as Tata Motors, Kia, Toyota, Honda, MG Motors, and Skoda to obtain a broader view of the Indian automobile market.

2. Real-Time Data Integration

Future versions of the project can use live or periodically updated sales data from reliable online sources, enabling real-time market monitoring and more accurate trend analysis.

3. Advanced Predictive Modelling

More sophisticated machine learning algorithms such as XGBoost, Gradient Boosting, Support Vector Regression, and Neural Networks can be implemented to improve forecasting accuracy and predictive performance.

4. Electric Vehicle Market Analytics

With the rapid growth of electric mobility in India, the project can be expanded to analyze EV sales trends, charging infrastructure impact, battery preferences, and customer adoption behaviour.

5. Regional and Demographic Analysis

State-wise, city-wise, or demographic-based datasets can be integrated to study regional demand patterns, income-based preferences, and customer segmentation.

6. Customer Sentiment Analysis

Online reviews, feedback, and social media data can be analyzed using Natural Language Processing (NLP) to understand customer satisfaction, brand perception, and market sentiment.

7. Enhanced Interactive Dashboard

The **AutoIntel Dashboard** can be upgraded into a fully dynamic web platform with filters, downloadable reports, live charts, cloud database support, and mobile responsiveness.

8. Decision Support for Industry

The project can be developed into a business intelligence system for automobile companies to assist in pricing strategy, production planning, inventory management, and future demand forecasting.

9. Sustainability and Policy Analysis

Future studies may also include the impact of fuel prices, environmental regulations, taxation policies, and government incentives on automobile sales trends.

10. International Market Comparison

The framework of this project can be extended to compare Indian automobile trends with global markets, enabling wider economic and industrial insights.

Thus, the project not only achieves its present objectives but also offers significant potential for future expansion as an advanced automobile analytics and forecasting platform.